

Signals & Systems (EE504)
Pre-Reading Material – Lecture 7

Energy & Power Signal
Estimated Reading Time: 20–25 minutes

Why This Topic Matters

Signals can be studied not only in the time domain but also in terms of their energy, power, and frequency content. Understanding **energy spectral density, power spectral density, and correlation** helps analyze how signals behave and how similar signals are to one another.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Distinguish between **energy signals and power signals**.
- Explain the concept of **energy spectral density (ESD)**.
- Explain the concept of **power spectral density (PSD)**.
- Understand **autocorrelation and cross-correlation**.
- State the basic idea of the **Wiener–Khinchin theorem**.
- Relate signal energy and power to their frequency-domain representations.

Activate Your Prior Knowledge

1. Energy vs Power

A signal may exist for a finite duration or continue indefinitely.

Think: How would you distinguish a signal that has finite total energy from one that has finite average power?

2. Frequency Content

A signal can contain components at different frequencies.



Think: How could we determine how the signal's energy or power is distributed across frequency?

3. Comparing Signals

Suppose two signals appear at different times but have similar patterns.

Think: How could we measure the similarity between these two signals?

Prerequisite Concepts

You should be familiar with:

- Continuous-time signals
- Fourier series
- Fourier transform
- Average signal power
- Basic integration
- Complex numbers

Core Concepts

1. Energy of a Signal

The total energy of a continuous-time signal is obtained by integrating the squared magnitude of the signal over all time. An energy signal has **finite, non-zero total energy**.

2. Power of a Signal

A power signal has infinite total energy but finite, non-zero average power. Average power is calculated using the long-term average of the squared magnitude of the signal.

3. Energy Signals vs Power Signals

An **energy signal** has finite energy and zero average power, while a **power signal** has finite average power and generally infinite energy. Periodic signals are common examples of power signals.

4. Parseval's Theorem

Parseval's theorem relates the total energy of a signal in the time domain to its frequency-domain representation. It states that signal energy can be calculated by integrating the squared magnitude of its Fourier transform.

5. Energy Spectral Density (ESD)

Energy spectral density describes how the energy of a finite-energy signal is distributed over frequency. It is given by the squared magnitude of the Fourier transform.

6. Power Spectral Density (PSD)

Power spectral density describes how the average power of a power signal is distributed over frequency. For a power signal, the PSD is related to the Fourier transform of its autocorrelation function.

7. Correlation

Correlation measures the similarity or dependence between signals. **Cross-correlation** compares two different signals, while **autocorrelation** measures the similarity of a signal with a shifted version of itself.

8. Autocorrelation

Autocorrelation measures how similar a signal is to itself at different time shifts. For a non-zero finite-energy signal, autocorrelation has its largest value at zero shift.

Prepared by: Dr. Rajesh Kumar

Signals & Systems (EE504 (B)) | AY 2026–27

Course Repository: <https://github.com/rajesh101191/Signals-Systems-PIEMR-2026-EE504>



9. Cross-Correlation

Cross-correlation measures the similarity between two different signals as one signal is shifted relative to the other. It can be used to detect whether a particular signal pattern occurs within another signal.

10. Wiener–Khinchin Theorem

The Wiener–Khinchin theorem establishes a relationship between correlation and spectral density. For a finite-energy signal, the Fourier transform of its autocorrelation gives its **energy spectral density**.

11. Signal Through a System

When a signal passes through a system with transfer function $H(\omega)$, its energy spectral density is modified according to the magnitude squared of the system's frequency response. Thus, $E_g(\omega) = |H(\omega)|^2 E_f(\omega)$.

12. Applications of Correlation

Correlation can be used for practical signal detection and synchronization. Applications include **ECG heartbeat synchronization** and detecting repeated patterns in signals, such as signals studied in SETI.

Guided Reading Questions

1. What is the difference between an energy signal and a power signal?
2. Why do periodic signals generally belong to the power-signal category?
3. What does energy spectral density represent?
4. What does power spectral density represent?
5. What is the difference between autocorrelation and cross-correlation?
6. Why is autocorrelation generally maximum at zero shift?

7. How is correlation useful for detecting patterns in signals?
8. What is the basic relationship established by the Wiener–Khinchin theorem?

Reflection Activity

Write 100–150 words explaining how **energy/power spectra and correlation** provide information that may not be obvious from the time-domain representation of a signal.

Self-Assessment

1. An energy signal has finite total energy. **(True/False)**
2. A power signal has finite, non-zero average power. **(True/False)**
3. Energy spectral density is represented by _____.
4. The similarity of a signal with a shifted version of itself is measured using _____.
5. The similarity between two different signals is measured using _____.
6. The autocorrelation of a non-zero finite-energy signal is maximum at _____.
7. PSD stands for _____.
8. ESD stands for _____.
9. The Fourier transform of the autocorrelation of a finite-energy signal gives its _____.
10. For a system with transfer function $H(\omega)$, the output ESD is multiplied by _____.

Key Takeaway

Energy and power spectra describe how a signal's energy or average power is distributed across frequency, while **correlation** measures similarity and dependence between signals. Together, these concepts provide powerful tools for analyzing signals in both the time and frequency domains.

